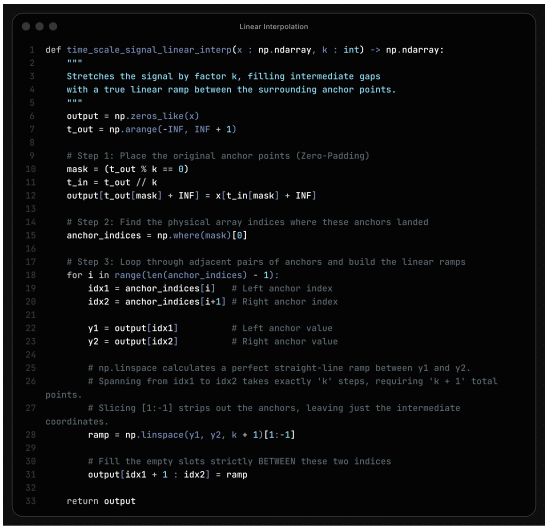
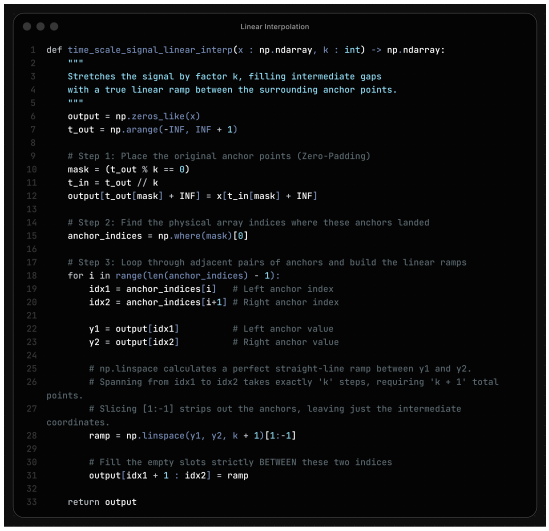
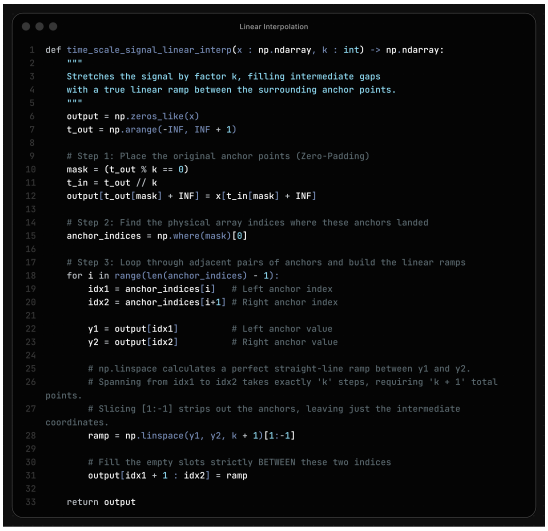


Figure 1: The reference waveform



(a) figure left



(b) figure right

Figure 2: whole figure

Ex 05

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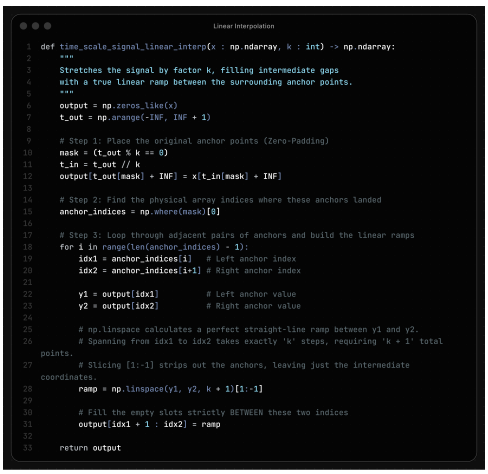

Figure 3: A wrapped figure n the right

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Figure 4 sits on the right while this paragraph flows around it. Lorem ipsum ... (enough text to fill the figures height)

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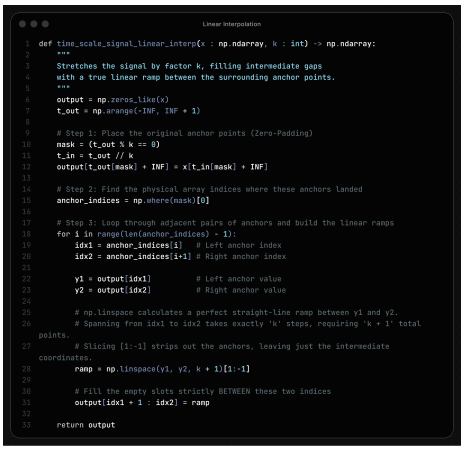

Figure 4: A wrapped figure on the right.

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Ex 08

(a) Framed (b) Fixed height (c) Cropped (d) Stretched

Figure 5: Four ways to control the size and extent of an image.

1 Ex 09

Linear Interpolation

```python
def time_scale_signal_linear_interp(x : np.ndarray, k : int) -> np.ndarray:
    """
    Stretches the signal by factor k, filling intermediate gaps
    with a true linear ramp between the surrounding anchor points.
    """
    output = np.zeros_like(x)
    t_out = np.arange(-INF, INF + 1)

    # Step 1: Place the original anchor points (Zero-Padding)
    mask = (t_out % k == 0)
    t_in = t_out // k
    output[t_out[mask] + INF] = x[t_in[mask] + INF]

    # Step 2: Find the physical array indices where these anchors landed
    anchor_indices = np.where(mask)[0]

    # Step 3: Loop through adjacent pairs of anchors and build the linear ramps
    for i in range(len(anchor_indices) - 1):
        idx1 = anchor_indices[i]   # Left anchor index
        idx2 = anchor_indices[i+1] # Right anchor index

        y1 = output[idx1]          # Left anchor value
        y2 = output[idx2]          # Right anchor value

        # np.linspace calculates a perfect straight-line ramp between y1 and y2.
        # Spanning from idx1 to idx2 takes exactly 'k' steps, requiring 'k + 1' total points.
        # Slicing [1:-1] strips out the anchors, leaving just the intermediate coordinates.
        ramp = np.linspace(y1, y2, k + 1)[1:-1]

        # Fill the empty slots strictly BETWEEN these two indices
        output[idx1 + 1 : idx2] = ramp

    return output
```

Figure 6: Measured step response.

Table 1: Extracted parameters.

Quantity	**Value**
Time constant τ	10.0 ms
Rise time t_r	22.0 ms
Overshoot	4.3%